

Q.1. Which scheduling algorithm allocates the CPU first to the process that requests the CPU first.?

- 1) first-come, first-served scheduling**
- 2) shortest job scheduling**
- 3) priority scheduling**
- 4) All of these**
- 5) None of the mentioned**

Q.2. In priority scheduling algorithm.?

- 1) CPU is allocated to the process with highest priority**
- 2) CPU is allocated to the process with lowest priority**
- 3) Equal priority processes can not be scheduled**
- 4) All of these**
- 5) None of the mentioned**

Q.3. In priority scheduling algorithm, when a process arrives at the ready queue, its priority is compared with the priority of.?

- 1) all process**
- 2) currently running process**
- 3) parent process**
- 4) init process**
- 5) All of these**

Q.4. Which algorithm is defined in Time quantum?

- 1) shortest job scheduling algorithm**
- 2) round robin scheduling algorithm**
- 3) priority scheduling algorithm**
- 4) multilevel queue scheduling algorithm**
- 5) All of these**

Q.5. Which of the following scheduling algorithm is provably optimal, in that it gives the minimum average waiting time for a given set of processes.?

- 1) Round Robin**
- 2) FCFS**
- 3) Priority**
- 4) SJFS**
- 5) All of these**

Q.6. The problem of indefinite blockage of low-priority jobs in general priority scheduling algorithm can be solved using.?

- 1) Dirty Bit**
- 2) Compaction**
- 3) Aging**
- 4) Swapping**
- 5) All of these**

Q.7. The real-time operating system, which of the following is the most suitable scheduling scheme.?

- 1) Round Robin**
- 2) FCFS**
- 3) Pre-emptive**
- 4) Random Scheduling**
- 5) All of these**

Q.8. Round Robin Scheduling is most suitable for.?

- 1) Batch OS**
- 2) Hard RTOS**
- 3) Soft RTOS**
- 4) Time Sharing OS**
- 5) None of these**

Q.9. Whenever the CPU becomes idle, the operating system must select one of the processes in the ____ to be executed.?

- 1) Execution Queue**
- 2) Process Queue**
- 3) Waiting Queue**
- 4) Ready Queue**
- 5) None of these**

Q.11. A major problem with priority scheduling is.?

- 1) Definite blocking**
- 2) Starvation**
- 3) Low priority**
- 4) All of these**
- 5) None of the above**

Q.12. Round robin scheduling is essentially the preemptive version of.?

- 1) FIFO/FCFS**
- 2) Shortest job first**
- 3) Shortest remaining**
- 4) Longest time first**
- 5) All of these**

Q.13. FIFO scheduling is.?

- 1) Preemptive Scheduling**
- 2) Non Preemptive Scheduling**
- 3) Deadline Scheduling**
- 4) Fair share scheduling**
- 5) All of these**

Q.14. Which scheduling policy is most suitable for time shared operating system.?

- 1) Shortest job first**
- 2) FCFS**
- 3) LCFS**
- 4) Round robin**
- 5) None of these**

Q.15. Which among following scheduling algorithms give minimum average waiting time.?

- 1) FCFS**
- 2) SJF**
- 3) Round robin**
- 4) On priority**
- 5) None of these**

Q.16. In which scheduling policies, context switching never takes place.?

- 1) FCFS**
- 2) Round robin**
- 3) Shortest job first**
- 4) Pre-emptive**
- 5) All of these**

Q.17. Shortest Job First executes first the job .?

- 1) With the least processor needs**
- 2) That first entered the queue**
- 3) That has been in the queue for the longest**
- 4) That last entered the queue**
- 5) None of these**

Q.18. Waiting time is.?

- 1) The total time in the blocked and waiting queues**
- 2) The total time spent in the ready queue**
- 3) The total time spent in the running queue**
- 4) The total time from the completion till the submission of a process**
- 5) None of these**

Q.19. The most optimal scheduling algorithm is.?

- 1) FCFS – First come First served**
- 2) SJF – Shortest Job First**
- 3) RR – Round Robin**
- 4) All of these**
- 5) None of these**

Q.20. Aging is.?

- 1) Keeping track of cache contents**
- 2) Keeping track of what pages are currently residing in memory**
- 3) Keeping track of how many times a given page is referenced**
- 4) Increasing the priority of jobs to ensure termination in a finite time**
- 5) None of these**

Q.21. In Shortest-Job-First Scheduling If the next CPU bursts of two processes are the same, _____ scheduling is used to break the tie.?

- 1) FCFS**
- 2) SJF**
- 3) Round – robin**
- 4) Priority**
- 5) All of these**

Q.22. A _____ process is moved to the ready state when its time allocation expires.?

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- 1) Blocked**
- 2) New**
- 3) Running**
- 4) Suspend**
- 5) All of these**

Q.23. What name is given to a program in execution.?

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- 1) Process**
- 2) Data Load**
- 3) Program**
- 4) Mutex**
- 5) All of these**

Q.24. Which of the following is not a valid Process State.?

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- 1) Ready**
- 2) Hold**
- 3) New**
- 4) Running**
- 5) All of these**

Q.25. Which of the following is not a valid Process State.?

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- 1) Ready**
- 2) Waiting**
- 3) Running**
- 4) Starving**
- 5) All of these**

Q.26. In which of the following state, the process is waiting for processor.?

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- 1) New**
- 2) Waiting**
- 3) Ready**
- 4) Running**
- 5) All of these**

Q.27. Which of the following states of process in operating system is defined incorrectly? **Arora Educator**



- 1) New : The process is being created**
- 2) Running : Instructions are being executed**
- 3) Waiting : Process is waiting to be assigned to a processor**
- 4) Terminated : The process has finished execution**
- 5) All of these**

Q.28. The systems which allow only one process execution at a time, are called.?

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- 1) uniprogramming systems**
- 2) uniprocessing systems**
- 3) unitasking systems**
- 4) None of the mentioned**
- 5) All of these**

Q.29. In Unix, Which system call creates the new process.?

Arora Edu



- 1) fork**
- 2) create**
- 3) New**
- 4) All of these**
- 5) None of the mentioned**

Q.30. A process can be terminated due to.?

33

- 1) normal exit**
- 2) fatal error**
- 3) killed by another process**
- 4) All of these**
- 5) None of these**

Q.31. What is the ready state of a process.? Arora

- 1) When process is scheduled to run after some execution**
- 2) When process is unable to run until some task has been completed**
- 3) When process is using the CPU**
- 4) All of these**
- 5) None of the mentioned**

Q.32. What is interprocess communication?

- 1) communication within the process**
- 2) communication between two process**
- 3) communication between two threads of same process**
- 4) All of these**
- 5) None of the mentioned**

Q.33. A set of processes is deadlock if.?

- 1) each process is blocked and will remain so forever**
- 2) each process is terminated**
- 3) all processes are trying to kill each other**
- 4) All of these**
- 5) None of the mentioned**

Q.34. How many states can be processed in Operating System.?

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- 1) 5
- 2) 4
- 3) 3
- 4) 1

Q.35. PCB stands for.?

- 1) Process Current Block**
- 2) Parent Control Block**
- 3) Parent Current Block**
- 4) Process Control Block**
- 5) None of these**

Q.36. A process will _____ while it is running, either because it has completed or because of some fatal fault condition.?

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- 1) Suspend**
- 2) Terminate**
- 3) Blocked**
- 4) Ready**
- 5) All of these**

Q.37. In _____ state, the process is in secondary memory but it is available for execution as soon as it is loaded into main memory..?

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- 1) Ready**
- 2) Blocked**
- 3) Blocked/Suspend**
- 4) Ready/Suspended**
- 5) All of these**

Q.38. The ready queue is generally stored as a.?

Arora

- 1) Array**
- 2) Stack**
- 3) Linked List**
- 4) All of these**
- 5) None of these**



Q.39. At the ready queue when a process arrives In priority scheduling algorithm, the priority of this process is compared with the priority of.?

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- 1) currently running process**
- 2) parent process**
- 3) all process**
- 4) init process**
- 5) All of these**

Q.40. The processes that are inhabited in main memory and are ready and waiting to execute and remained on a list called.?

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- 1) process queue**
- 2) execution queue**
- 3) job queue**
- 4) ready queue**
- 5) All of these**

Q.41. Second Stage of Process life Cycle is.?

A

- 1) Ready**
- 2) New**
- 3) Running**
- 4) Wait**
- 5) All of these**

Q.42. Which module gives control of the CPU to the process selected by the short-term scheduler.?

- 1) dispatcher**
- 2) interrupt**
- 3) Scheduler**
- 4) All of these**
- 5) None of the mentioned**

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Q.43. The interval from the time of submission of a process to the time of completion is termed as.?

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- 1) waiting time**
- 2) turnaround time**
- 3) response time**
- 4) Throughput**
- 5) All of these**

Q.44. Whenever the CPU becomes idle, the operating system must select one of the processes in the ____ to be executed.?

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- 1) Execution Queue**
- 2) Process Queue**
- 3) Waiting Queue**
- 4) Ready Queue**
- 5) All of these**

Q.45. CPU performance is measured through.?

- 1) Throughput**
- 2) MHz**
- 3) Flaps**
- 4) All of these**
- 5) None of the above**

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Q.46. In the multi-programming environment, the main memory consisting of ____ number of process.?

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- 1) Greater than 100**
- 2) Only one**
- 3) Greater than 50**
- 4) More than one**
- 5) None of these**

Q.47. What is the minimum number of processes that can have deadlock amongst themselves? **Arora Educator**



- 1) 1
- 2) 2
- 3) 3
- 4) 4
- 5) 5

Q.48. The Banker's Algorithm in an Operating System is used for.?

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- 1) Deadlock Recovery**
- 2) Interprocess Communication**
- 3) Mutual Exclusion**
- 4) Deadlock Avoidance**
- 5) All of these**

Q.49. A process said to be in _____ state, if it was waiting for an event that will never occur?

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- 1) Safe
- 2) Unsafe
- 3) Starvation
- 4) Deadlock
- 5) Context Switching

Q.50. Which of the following is NOT the method for handling deadlock.?

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- 1) Detection**
- 2) Recovery**
- 3) Prevention**
- 4) Deletion**
- 5) All of these**

Q.51. The circular wait condition can be prevented by.?

- 1) defining a linear ordering of resource types**
- 2) using thread**
- 3) using pipes**
- 4) all of the mentioned**
- 5) None of these**

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Q.52. What is the drawback of banker's algorithm.?

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- 1) in advance processes rarely know how much resource they will need
- 2) the number of processes changes as time progresses
- 3) resource once available can disappear
- 4) All of the mentioned
- 5) None of these

Q.53. It is the problem that occurs when high priority processes keep executing and low priority processes get blocked for indefinite time.?

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- 1) Deadlock**
- 2) Starvation**
- 3) Inversion**
- 4) Aging**
- 5) All of these**

Q.54. To avoid deadlock.?

- 1) there must be a fixed number of resources to allocate**
- 2) resource allocation must be done only once**
- 3) all deadlocked processes must be aborted**
- 4) inversion technique can be used**
- 5) All of these**

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Q.55. When a process is executed in a critical section, then no other process can be executed in their critical section.?

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- 1) Mutual Exclusion**
- 2) Hold & Wait**
- 3) No Preemption**
- 4) Circular Wait**
- 5) All of these**

Q.56. Deadlock Condition in which a process is currently holding at least one resource and requesting additional resources is called .?

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- 1) Mutual Exclusion**
- 2) Hold & Wait**
- 3) No Preemption**
- 4) Circular Wait**
- 5) All of these**

Q.57. Deadlock Condition in which a resource can be released only voluntarily by the process holding is called .?

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- 1) Mutual Exclusion**
- 2) Hold & Wait**
- 3) No Preemption**
- 4) Circular Wait**
- 5) All of these**

Q.58. Deadlock Condition in which each process must be waiting for a resource which is being held by another process, which in turn is waiting for the first process is called .?

- 1) Mutual Exclusion**
- 2) Hold & Wait**
- 3) No Preemption**
- 4) Circular Wait**
- 5) All of these**

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Q.59. An operating system has N resources of the same type. These resources are shared by 4 processes a, b, c, and d which have peak demands of 3, 5, 8, and 7 respectively. For what value of N deadlock will not occur?

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- 1) 23**
- 2) 4**
- 3) 20**
- 4) 21**
- 5) 10**

Q.60. Which of the following is not a type of Interrupt in OS.?

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- 1) Hardware Interrupt**
- 2) Software Interrupt**
- 3) Memory Interrupt**
- 4) More than one of the above**
- 5) None of these**

Q.61. An interrupt that can be temporarily ignored is.?

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- 1) Vectored interrupt**
- 2) Non-maskable interrupt**
- 3) Maskable interrupt**
- 4) High priority interrupt**
- 5) All of these**

Q.62. The interrupt-request line is a part of the _____.

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- 1) Data line**
- 2) Control line**
- 3) Address line**
- 4) All of these**
- 5) None of the mentioned**

Q.63. The return address from the interrupt-service routine is stored on the?

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- 1) System heap**
- 2) Processor register**
- 3) Processor stack**
- 4) Memory**
- 5) More than one of the above**

Q.64. The signal sent to the device from the processor to the device after receiving an interrupt is?

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- 1) Interrupt-acknowledge**
- 2) Return signal**
- 3) Service signal**
- 4) Permission signal**
- 5) All of these**

Q.65. The time between the receiver of an interrupt and its service is?

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- 1) Interrupt delay**
- 2) Interrupt latency**
- 3) Cycle time**
- 4) Switching time**
- 5) All of these**

Q.66. Interrupts form an important part of _____ systems.?

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- 1) Batch processing**
- 2) Multitasking**
- 3) Real-time processing**
- 4) Multi-user**
- 5) All of these**

Q.67. Which interrupt has the highest priority?

- 1) INTR
- 2) TRAP
- 3) RST 6.5
- 4) All of these
- 5) None of these

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Q.68. Which of the following is software interrupts?

- 1) RST 0-7**
- 2) RST 5.5**
- 3) RST 6.5**
- 4) RST 7.5**
- 5) TRAP**

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**Q.69. _____interrupt used for emergency purpose
e.g power failure, smoke detector etc.?**

72

- 1) Non maskable**
- 2) Maskable**
- 3) Executable**
- 4) All of these**
- 5) None of these**

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Q.70. _____ interrupts used to interface with peripheral device.?

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- 1) Non maskable**
- 2) Maskable**
- 3) Executable**
- 4) All of these**
- 5) None of these**

Q.71. This Interrupt is ignored by the instructions of CPU.?

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- 1) Non maskable**
- 2) Maskable**
- 3) Executable**
- 4) All of these**
- 5) None of these**

Q.72. This Interrupt cannot ignored by the instructions of CPU.?

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- 1) Non maskable**
- 2) Maskable**
- 3) Executable**
- 4) All of these**
- 5) None of these**

Q.73. _____ is a signal that causes the operating system (OS) to stop and determine what to do next.?

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- 1) Abort**
- 2) Kill**
- 3) Interrupt**
- 4) All of these**
- 5) None of these**

Q.74. Software interrupts are triggered by software instructions.?

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- 1) System Calls**
- 2) Exceptions**
- 3) Breakpoints**
- 4) All of these**
- 5) None of these**

Q.75. Which of the following file system commonly used in Operating System.?

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- 1) NTFS**
- 2) FAT32**
- 3) HFS+**
- 4) More than one of the above**
- 5) None of these**

Q.116. Each entry in a segment table has a.?

Arora

- 1) Segment base**
- 2) Segment peak**
- 3) Segment value**
- 4) All of these**
- 5) None of these**

Q.117. The segment limit contains the.?

Arora

- 1) Starting logical address of the process**
- 2) Starting physical address of the segment in memory**
- 3) Segment length**
- 4) All of these**
- 5) None of these**

Q.118. Which of the following is NOT a memory management technique used by operating systems.?

- 1) Paging**
- 2) Segmentation**
- 3) Fragmentation**
- 4) More than one of the above**
- 5) None of the above**

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Q.119. Which one of the following is the address generated by CPU.?

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- 1) physical address**
- 2) absolute address**
- 3) logical address**
- 4) All of these**
- 5) None of these**

Q.120. Run time mapping from virtual to physical address is done by.?

Arora Educator



- 1) Memory management unit**
- 2) CPU**
- 3) PCI**
- 4) All of these**
- 5) None of these**

Q.121. Program always deals with.?

- 1) logical address**
- 2) absolute address**
- 3) physical address**
- 4) relative address**
- 5) All of these**

Q.122. Operating System maintains the page table for.?

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- 1) each process**
- 2) each thread**
- 3) each instruction**
- 4) each address**
- 5) All of these**

Q.123. In which of the below phenomenon memory is wasted.?

Arora Educator



- 1) Compaction**
- 2) Fragmentation**
- 3) Multiplexing**
- 4) Paging**
- 5) Thrashing**

Q.124.Which of the following is a type of operating System.?

Arora Educator



- 1) Real-time operating system**
- 2) Embedded operating system**
- 3) Network operating system**
- 4) More than one of the above**
- 5) None of these**

Q.124.Which of the following is a type of operating System.?

Arora Educator



- 1) Real-time operating system**
- 2) Embedded operating system**
- 3) Network operating system**
- 4) More than one of the above**
- 5) None of these**

Q.125. Thread shares with other threads belonging to the same process its.?

Arora Educator



- 1) thread ID**
- 2) program counter**
- 3) code section and data section**
- 4) More than one of the above**
- 5) None of the above**

Q.125. Thread shares with other threads belonging to the same process its.?

Arora Educator



- 1) thread ID**
- 2) program counter**
- 3) code section and data section**
- 4) More than one of the above**
- 5) None of the above**

Q.126. Consider a disk queue with requests for I/O to blocks on cylinders.

98 183 37 122 14 124 65 67

Considering FCFS (first cum first served) scheduling, the total number of head movements is, if the disk head is initially at 53 is.?

- 1) 600**
- 2) 620**
- 3) 640**
- 4) 639**
- 5) All of these**

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Q.127. Semaphore is a/an _____ to solve the critical section problem.?

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- 1) Hardware for a system**
- 2) Special program for a system**
- 3) Integer Variable**
- 4) All of these**
- 5) None of these**

Q.128. What are the two atomic operations permissible on semaphores.?

Arora Educator



- 1) wait and signal
- 2) stop and wait
- 3) hold and signal
- 4) All of these
- 5) None of these

Q.129. The wait operation of the semaphore basically works on the basic _____ system call.? Arora Educator



- 1) stop()
- 2) block()
- 3) hold()
- 4) wait()
- 5) All of these

Q.130. The signal operation of the semaphore basically works on the basic _____ system call.? Arora Educator



- 1) `continue()`
- 2) `wakeup()`
- 3) `getup()`
- 4) `start()`
- 5) None of these

Q.131. In order to allow only one process to enter its critical section, binary semaphore are initialized to.?

- 1) 0**
- 2) 1**
- 3) 2**
- 4) 3**
- 5) 4**

Arora Educator



Q.132. Semaphores are used to solve the problem of.?

- 1) Mutual Exclusion**
- 2) Process Synchronization**
- 3) Hold & Wait**
- 4) Both 1 & 2**
- 5) Both 2 & 3**

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Q.133. What are the two types of Semaphore.?

Arora Educator



- 1) Digital Semaphores and Binary Semaphores**
- 2) Analog Semaphores and Octal Semaphores**
- 3) Counting Semaphores and Binary Semaphores**
- 4) Critical Semaphores and System Semaphores**
- 5) None of these**

Q.134. Which of the following page replacement algorithms suffers from Belady's Anomaly? **Arora Educator**



- 1) Optimal replacement
- 2) LRU
- 3) FIFO
- 4) Both optimal replacement and FIFO
- 5) None of these

Q.135. Increasing the RAM of a computer typically improves performance because.?

Arora Educato



- 1) Virtual memory increases**
- 2) Larger RAMs are faster**
- 3) Fewer page faults occur**
- 4) All of these**
- 5) None of these**

Q.136. In a FIFO algorithm, when a page is to be replaced, which of the following page is chosen.

Arora Educator



- 1) Oldest page**
- 2) Newest page**
- 3) Frequently occurred page in the future**
- 4) Not frequently occurred page in the future**
- 5) All of these**
- 6) None of these**

Q.137. Operating system supports different page replacement policy. From the given below option which is not a valid page replacement policy.?

Arora Educator



- 1) Least Recently Used**
- 2) First in first out**
- 3) Round robin**
- 4) Optimal page replacement policy**
- 5) All of these**

Q.138. The optimal page replacement algorithm will select the page that.?

Arora Educator



- 1) has been used least number of times**
- 2) has been used most number of times**
- 3) has been used for the longest time in the past**
- 4) will not be used for the longest time in future**
- 5) All of these**

Q.155. The _____ technique remembers which entry in the free list was used to make the last allocation.?

Arora Educator



- 1) First Fit technique**
- 2) Best-Fit technique**
- 3) Next-Fit technique**
- 4) Good-Fit technique**
- 5) All of these**

Q.139. 'LRU' page replacement policy is.? **Arora Educator**



- 1) Last Replaced Unit**
- 2) Last Restored Unit**
- 3) Least Recently Used**
- 4) Least Required Unit**
- 5) None of these**

Q.140. Which amongst the following is not a valid page replacement policy .?

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- 1) LRU policy (Least Recently Used)**
- 2) FIFO policy (First in first out)**
- 3) RU policy (Recurrently used)**
- 4) Optimal page replacement policy**
- 5) None of these**

Q.141. Virtual memory uses disk space as an extension of.?

Arora Educator



- 1) Secondary storage**
- 2) Main memory**
- 3) Tertiary storage**
- 4) All of these**
- 5) None of these**

Q.142. Full form of TLB is _____.

- 1) Transfer Lookaside Buffer**
- 2) Translation Lookinside Buffer**
- 3) Translation Lookaside Buffer**
- 4) All of these**
- 5) None of these**

Q.143. In segmentation, each address is specified by.?

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- 1) a segment number & offset**
- 2) an offset & value**
- 3) a value & segment number**
- 4) a key & value**
- 5) All of these**

Q.144. In internal fragmentation, memory is internal to a partition and.?

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- 1) is being used
- 2) is not being used
- 3) is always used
- 4) None of these
- 5) All of these

Q.145. A solution to the problem of external fragmentation is.?

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- 1) compaction**
- 2) larger memory space**
- 3) smaller memory space**
- 4) None of these**
- 5) All of these**

Q.146. Memory fragmentation results in_____.

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- 1) stack overflow**
- 2) page faults**
- 3) Better Utilization of memory**
- 4) poor utilization of memory**
- 5) None of these**

Q.147. Which conditions must be satisfied to solve a critical section problem.?

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- 1) Bounded Waiting**
- 2) Progress**
- 3) Mutual Exclusion**
- 4) All of these**
- 5) None of these**

Q.148. The first fit, best fit and worst fit are strategies to select a _____.?

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- 1) process from a queue to put in memory**
- 2) processor to run the next process**
- 3) free hole from a set of available holes**
- 4) All of these**
- 5) None of these**

Q.149. _____ is generally faster than _____ and _____.

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- 1) first fit, best fit, worst fit
- 2) best fit, first fit, worst fit
- 3) worst fit, best fit, first fit
- 4) All of these
- 5) None of these

Q.150. External fragmentation will not occur when?

Arora Educator



- 1) first fit is used**
- 2) best fit is used**
- 3) worst fit is used**
- 4) no matter which algorithm is used, it will always occur**
- 5) None of these**

Q.151. Amongst following pairs, which combination is correct for dynamic memory allocation systems?

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- 1) Best Fit: Big holes**
- 2) First Fit: Faster than Best and Worst Fit**
- 3) Worst Fit: Small holes**
- 4) All of these**
- 5) None of the above**

Q.152. Which of the following fragmentation types occurs when free memory blocks are scattered throughout the memory space.?

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- 1) Internal Fragmentation**
- 2) External Fragmentation**
- 3) Contiguous Fragmentation**
- 4) Paging Fragmentation**
- 5) All of these**

Q.116. Each entry in a segment table has a.?

Arora

- 1) Segment base**
- 2) Segment peak**
- 3) Segment value**
- 4) All of these**
- 5) None of these**

Q.117. The segment limit contains the.?

Arora

- 1) Starting logical address of the process**
- 2) Starting physical address of the segment in memory**
- 3) Segment length**
- 4) All of these**
- 5) None of these**

Q.118. Which of the following is NOT a memory management technique used by operating systems.?

- 1) Paging**
- 2) Segmentation**
- 3) Fragmentation**
- 4) More than one of the above**
- 5) None of the above**

Arora Educator



Q.119. Which one of the following is the address generated by CPU.?

Arora Educator



- 1) physical address**
- 2) absolute address**
- 3) logical address**
- 4) All of these**
- 5) None of these**

Q.120. Run time mapping from virtual to physical address is done by.?

Arora Educator



- 1) Memory management unit**
- 2) CPU**
- 3) PCI**
- 4) All of these**
- 5) None of these**

Q.121. Program always deals with.?

- 1) logical address**
- 2) absolute address**
- 3) physical address**
- 4) relative address**
- 5) All of these**

Q.122. Operating System maintains the page table for.?

Arora Educator



- 1) each process**
- 2) each thread**
- 3) each instruction**
- 4) each address**
- 5) All of these**

Q.123. In which of the below phenomenon memory is wasted.?

Arora Educator



- 1) Compaction**
- 2) Fragmentation**
- 3) Multiplexing**
- 4) Paging**
- 5) Thrashing**

Q.124.Which of the following is a type of operating System.?

Arora Educator



- 1) Real-time operating system**
- 2) Embedded operating system**
- 3) Network operating system**
- 4) More than one of the above**
- 5) None of these**

Q.124.Which of the following is a type of operating System.?

Arora Educator



- 1) Real-time operating system**
- 2) Embedded operating system**
- 3) Network operating system**
- 4) More than one of the above**
- 5) None of these**

Q.125. Thread shares with other threads belonging to the same process its.?

Arora Educator



- 1) thread ID**
- 2) program counter**
- 3) code section and data section**
- 4) More than one of the above**
- 5) None of the above**

Q.125. Thread shares with other threads belonging to the same process its.?

Arora Educator



- 1) thread ID**
- 2) program counter**
- 3) code section and data section**
- 4) More than one of the above**
- 5) None of the above**

Q.126. Consider a disk queue with requests for I/O to blocks on cylinders.

98 183 37 122 14 124 65 67

Considering FCFS (first cum first served) scheduling, the total number of head movements is, if the disk head is initially at 53 is.?

- 1) 600**
- 2) 620**
- 3) 640**
- 4) 639**
- 5) All of these**

Arora Educator



Q.127. Semaphore is a/an _____ to solve the critical section problem.?

Arora Educator



- 1) Hardware for a system**
- 2) Special program for a system**
- 3) Integer Variable**
- 4) All of these**
- 5) None of these**

Q.128. What are the two atomic operations permissible on semaphores.?

Arora Educator



- 1) wait and signal
- 2) stop and wait
- 3) hold and signal
- 4) All of these
- 5) None of these

Q.129. The wait operation of the semaphore basically works on the basic _____ system call.? **Arora Educator**



- 1) stop()
- 2) block()
- 3) hold()
- 4) wait()
- 5) All of these

Q.130. The signal operation of the semaphore basically works on the basic _____ system call.? Arora Educator



- 1) continue()
- 2) wakeup()
- 3) getup()
- 4) start()
- 5) None of these

Q.131. In order to allow only one process to enter its critical section, binary semaphore are initialized to.?

- 1) 0**
- 2) 1**
- 3) 2**
- 4) 3**
- 5) 4**

Arora Educator



Q.132. Semaphores are used to solve the problem of.?

- 1) Mutual Exclusion**
- 2) Process Synchronization**
- 3) Hold & Wait**
- 4) Both 1 & 2**
- 5) Both 2 & 3**

Arora Educator



Q.133. What are the two types of Semaphore.?

Arora Educator



- 1) Digital Semaphores and Binary Semaphores**
- 2) Analog Semaphores and Octal Semaphores**
- 3) Counting Semaphores and Binary Semaphores**
- 4) Critical Semaphores and System Semaphores**
- 5) None of these**

Q.134. Which of the following page replacement algorithms suffers from Belady's Anomaly? **Arora Educator**



- 1) Optimal replacement
- 2) LRU
- 3) FIFO
- 4) Both optimal replacement and FIFO
- 5) None of these

Q.135. Increasing the RAM of a computer typically improves performance because.?

Arora Educato



- 1) Virtual memory increases**
- 2) Larger RAMs are faster**
- 3) Fewer page faults occur**
- 4) All of these**
- 5) None of these**

Q.136. In a FIFO algorithm, when a page is to be replaced, which of the following page is chosen.

Arora Educator



- 1) Oldest page**
- 2) Newest page**
- 3) Frequently occurred page in the future**
- 4) Not frequently occurred page in the future**
- 5) All of these**
- 6) None of these**

Q.137. Operating system supports different page replacement policy. From the given below option which is not a valid page replacement policy.?

Arora Educator



- 1) Least Recently Used**
- 2) First in first out**
- 3) Round robin**
- 4) Optimal page replacement policy**
- 5) All of these**

Q.138. The optimal page replacement algorithm will select the page that.?

Arora Educator



- 1) has been used least number of times**
- 2) has been used most number of times**
- 3) has been used for the longest time in the past**
- 4) will not be used for the longest time in future**
- 5) All of these**

Q.155. The _____ technique remembers which entry in the free list was used to make the last allocation.?

Arora Educator



- 1) First Fit technique**
- 2) Best-Fit technique**
- 3) Next-Fit technique**
- 4) Good-Fit technique**
- 5) All of these**

Q.139. 'LRU' page replacement policy is.? **Arora Educator**



- 1) Last Replaced Unit**
- 2) Last Restored Unit**
- 3) Least Recently Used**
- 4) Least Required Unit**
- 5) None of these**

Q.140. Which amongst the following is not a valid page replacement policy .?

Arora Educator



- 1) LRU policy (Least Recently Used)**
- 2) FIFO policy (First in first out)**
- 3) RU policy (Recurrently used)**
- 4) Optimal page replacement policy**
- 5) None of these**

Q.141. Virtual memory uses disk space as an extension of.?

Arora Educator



- 1) Secondary storage**
- 2) Main memory**
- 3) Tertiary storage**
- 4) All of these**
- 5) None of these**

Q.142. Full form of TLB is _____.

- 1) Transfer Lookaside Buffer**
- 2) Translation Lookinside Buffer**
- 3) Translation Lookaside Buffer**
- 4) All of these**
- 5) None of these**

Q.143. In segmentation, each address is specified by.?

Arora Educator



- 1) a segment number & offset**
- 2) an offset & value**
- 3) a value & segment number**
- 4) a key & value**
- 5) All of these**

Q.144. In internal fragmentation, memory is internal to a partition and.?

Arora Educator



- 1) is being used
- 2) is not being used
- 3) is always used
- 4) None of these
- 5) All of these

Q.145. A solution to the problem of external fragmentation is.?

Arora Educator



- 1) compaction**
- 2) larger memory space**
- 3) smaller memory space**
- 4) None of these**
- 5) All of these**

Q.146. Memory fragmentation results in_____.

Arora Educator



- 1) stack overflow**
- 2) page faults**
- 3) Better Utilization of memory**
- 4) poor utilization of memory**
- 5) None of these**

Q.147. Which conditions must be satisfied to solve a critical section problem.?

Arora Educator



- 1) Bounded Waiting**
- 2) Progress**
- 3) Mutual Exclusion**
- 4) All of these**
- 5) None of these**

Q.148. The first fit, best fit and worst fit are strategies to select a _____.?

Arora Educator



- 1) process from a queue to put in memory**
- 2) processor to run the next process**
- 3) free hole from a set of available holes**
- 4) All of these**
- 5) None of these**

Q.149. _____ is generally faster than _____ and _____.

Arora Educator



- 1) first fit, best fit, worst fit
- 2) best fit, first fit, worst fit
- 3) worst fit, best fit, first fit
- 4) All of these
- 5) None of these

Q.150. External fragmentation will not occur when?

Arora Educator



- 1) first fit is used**
- 2) best fit is used**
- 3) worst fit is used**
- 4) no matter which algorithm is used, it will always occur**
- 5) None of these**

Q.151. Amongst following pairs, which combination is correct for dynamic memory allocation systems?

Arora Educator



- 1) Best Fit: Big holes**
- 2) First Fit: Faster than Best and Worst Fit**
- 3) Worst Fit: Small holes**
- 4) All of these**
- 5) None of the above**

Q.152. Which of the following fragmentation types occurs when free memory blocks are scattered throughout the memory space.?

Arora Educator



- 1) Internal Fragmentation**
- 2) External Fragmentation**
- 3) Contiguous Fragmentation**
- 4) Paging Fragmentation**
- 5) All of these**

Q.153. To service a request for n bytes of memory, the _____ technique uses the first free memory area it can find whose size is $\geq n$ bytes.?

Arora Educator



- 1) First Fit technique**
- 2) Best-Fit technique**
- 3) Next-Fit technique**
- 4) Good-Fit technique**
- 5) All of these**

Q.154. The _____ technique uses the smallest free memory area with size $\geq n$ bytes.?

Arora Educator



- 1) First Fit technique**
- 2) Best-Fit technique**
- 3) Next-Fit technique**
- 4) Good-Fit technique**
- 5) All of these**

Q.155. The _____ technique remembers which entry in the free list was used to make the last allocation.?

Arora Educator



- 1) First Fit technique**
- 2) Best-Fit technique**
- 3) Next-Fit technique**
- 4) Good-Fit technique**
- 5) All of these**

Q.156. This is a modified version of 'first fit'.?

- 1) Best-Fit technique**
- 2) Next-Fit technique**
- 3) Good-Fit technique**
- 4) All of these**
- 5) None of these**

Arora Educator



Q.157. Consider a disk sequence with 100 cylinders.
The request to access the cylinder occur in the following
sequence :4, 34, 10, 7, 19, 73, 2, 15, 6, 20
Assuming that the head is currently at cylinder 50, what is the
time taken to satisfy all requests if it takes 2 ms to move from
one cylinder to adjacent one and **shortest seek time first policy**
is?

- 1) 190**
- 2) 238**
- 3) 233**
- 4) 276**
- 5) None of these**

Arora Educator



Q.158. Consider a disk sequence with 200 cylinders.

The request to access the cylinder occur in the following sequence :176, 79, 34, 60, 92, 11, 41, 114

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and **shortest seek time first policy**

- is?
- 1) 150
 - 2) 110
 - 3) 220
 - 4) 204
 - 5) 400

Arora Educator



Q.159. Consider a disk sequence with 200 cylinders.

The request to access the cylinder occur in the following sequence :82, 170, 43, 140, 24, 16, 190

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and **shortest seek time first policy**

- is?
- 1) 200
 - 2) 300
 - 3) 308
 - 4) 208
 - 5) None of these

Arora Educator



Q.160. Consider a disk queue with requests for I/O to blocks on cylinders.

98 183 37 122 14 124 65 67

Considering **FCFS** (first cum first served) scheduling, the total number of head movements is, if the disk head is initially at 53 is.?

- 1) 600
- 2) 620
- 3) 640
- 4) 639
- 5) All of these



Arora Educator

